Solving (sparse) system of linear equations (SLE) with successive overrelaxation method (SOR)

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1 Method:

The one-dimensional Poisson equation was discretized in order to get the an algebraic one. The specific distance Δ was set to define the computational domain.

$$\Delta = \frac{x_{max} - x_{min}}{n - 1} \tag{1}$$

The second derivative was replaced with a symmetric three-term second-order finite difference formula.

$$\frac{V_{i+1} - 2V_i + V_{i-1}}{\Delta^2} = -\rho_i \tag{2}$$

The result of this operation yields a sparse system of linear equations which can be expressed in matrix form:

$$A\vec{v} = \vec{\rho} \tag{3}$$

where A is symmetric tridiagonal matrix. Which diagonal elements are calculated as follow:

$$a_{i,i} = -\frac{2}{\Delta^2} \tag{4}$$

$$a_{i,\pm i} = \frac{1}{\Delta^2} \tag{5}$$

To minimize computational cost and skip matrix factorization, only non-zero matrix elements were taken into account. The tridiagonal matrix was stored using the Compressed Sparse Row (CSR) format, which requires three one-dimensional arrays: `a` for number of matrix elements, `col` for column indicies and `row` for global indices of first entries in rows.

The sparse system of linear equations was solved using the Successive Over-Relaxation (SOR) iterative method. In a single iteration, the entire vector v is updated by iterating all over its elements.

2 Tasks:

The practical part of the project consisted of several computational tasks designed to implement and test the SOR algorithm.

Firstly a project required writing a program to solve a system of linear equations (SLE) using the SOR method for a sparse matrix encoded in the CSR format. This included developing additional routines for calculating the scalar product of two vectors and performing matrix-vector multiplication.

In addition to this, the system must have been initialized with the appropriate task-provided values: $n = 500$, $\Delta = 1$, $itmax = 10^4$ and $\epsilon = 10^{-15}$. The right-handed side vector ρ was filled with values according to the formula:

$$\rho_i = \frac{1}{10}\left[\sin\left(\frac{2\pi \cdot i}{n-1}\right)\right]^{10}, \quad i = 0, 1, 2, \ldots, n-1 \tag{6}$$

Task required solving the SLE for the relaxation parameter $\omega = 1$ as well as $\omega = 1.5$, $\omega = 1.9$ and $\omega = 1.99$. During the iterative process, the iteration index, the norm of the approximated solution $||v||_2$ and the norm of residual vector $||r||_2$ were saved in different files for plotting. The data was collected to create a single comparative plot showing the residual vector norm changes depending on the iteration index for all ω values.

Finally some additional tests were performed for zero and randomly filled $\vec{v}$ along with $\epsilon = 10^{-8}$ and $\omega = 1.99$.

3 Results:

The initial calculations were performed using the relaxation parameter $\omega = 1$ and a constant initial approximated solution. Figure 1 and 2 present the spatial distribution of the right-hand-side vector (density source) $\vec{\rho}$ and the final approximated solution $\vec{v}$ depending on the spatial position x_i.

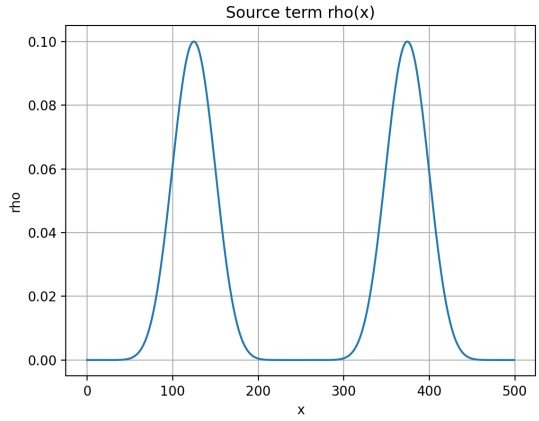


Figure 1: Distribution of the source vector $\vec{\rho}$ across the computational domain.

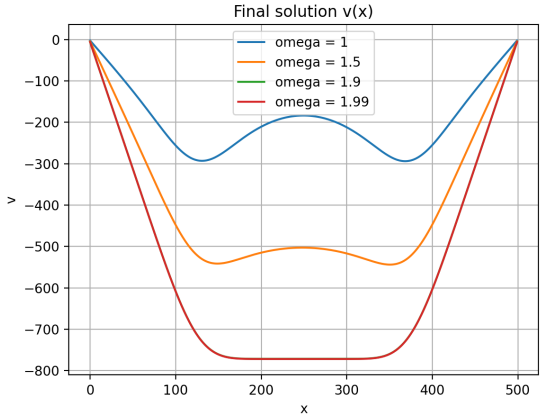


Figure 2: Distribution of the final solution vector $\vec{v}$ across the computational domain.

During the iterative process, the norm of the approximated solution $||\vec{v}||_2$ and the norm of the residual vector $||\vec{r}||_2$ were monitored. As shown in Figure 3 and 4, the residual norm steadily decreases, indicating the convergence of the method, while the solution norm stabilizes at a constant level as the system approaches the final solution.

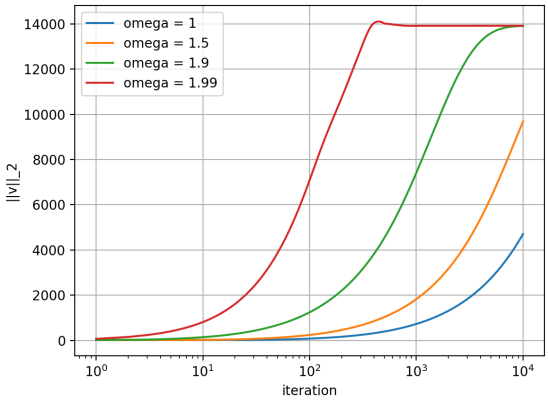


Figure 3: Convergence of the solution norm $||\vec{v}||_2$

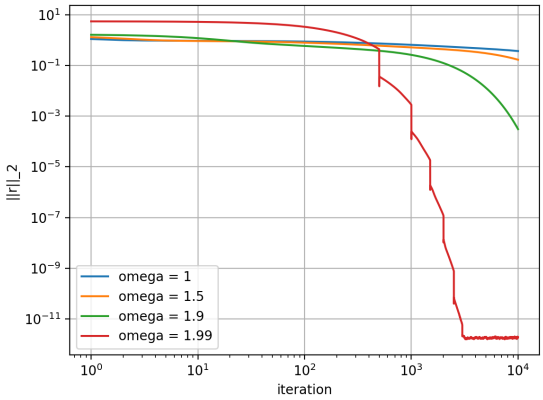


Figure 4: Residual vector norm $||\vec{r}||_2$

Figures 5 and 6 present how SOR dealt with randomly filled vector $\vec{v}$.

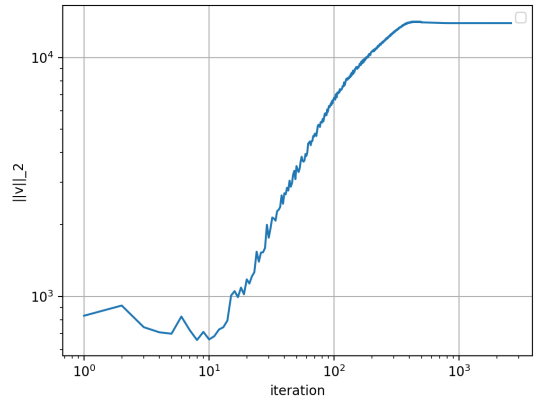


Figure 5: Residual vector norm for random $\vec{v}$ $||\vec{v}||_2$

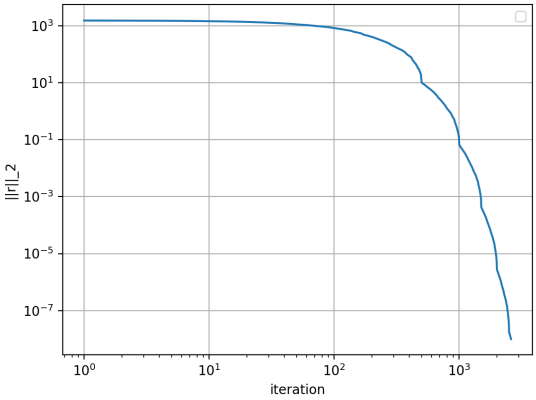


Figure 6: Residual vector norm for random $\vec{v}$ $||\vec{r}||_2$

4 Conclusions:

The results demonstrate that the SOR method is highly effective in solving sparse systems of linear equations, and its efficiency is strictly dependent on the choice of the ω parameter. Increasing the parameter from 1.0 to 1.99 significantly accelerated the convergence rate. The number of iterations required to reach the tolerance threshold dropped, proving that optimal over-relaxation drastically minimizes the computational cost.Finally, it is worth noting that the overall performance of the implemented solver was significantly enhanced by utilizing the Compressed Sparse Row (CSR) format for the system's tridiagonal matrix. By storing only the non-zero elements and avoiding operations on empty cells, the computational cost of the matrix-vector multiplication in each single iteration was heavily minimized. This approach proved to be a crucial factor in efficiently solving the sparse system of linear equations derived from the discretized Poisson equation.